



United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Exam Name: Final Term Exam Trimester: Fall 2025

Course Code: PMG 4101, Course Title: Project Management

Total Marks: 40

Duration: 2 hours

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

Answer all the questions.

Consider the following Scenario:

Universities in Bangladesh face significant challenges in managing research grants, faculty publications, and project compliance due to reliance on spreadsheets, emails, and fragmented departmental systems. Core activities such as proposal submission, funding approval, budget tracking, progress reporting, and publication management are largely manual, leading to missed deadlines, data inconsistency, limited transparency, and delayed reporting to funding agencies. To address these issues, **United International University (UIU)** has decided to implement a **Centralized Research Grant and Publication Management System** to streamline and automate these processes.

Your organization has been selected to design, develop, and deploy the system, which must support thousands of concurrent users during peak submission and reporting periods. The system will be used by faculty members, research assistants, department coordinators, reviewers, finance officers, and university administrators. The project team consists of 13 experienced software engineers handling system architecture, security, and integration, supported by 27 fresh graduates responsible for UI development, testing, user support, and data migration. University management expects regular milestone-based progress updates, demonstrations, and compliance reviews throughout the project lifecycle.

As the project manager, you are responsible for leading this initiative. The solution will be a **Web-based application with role-based access control**. Faculty members will submit research proposals, budget details, ethical clearance documents, and progress reports through structured forms. Reviewers will evaluate submissions and record approval decisions, while finance officers will monitor grant disbursements, expense claims, and remaining balances against approved budgets.

The system will maintain a centralized repository of research projects, funding sources, publications, and compliance documents, supporting multi-level approval workflows, revision cycles, deadline tracking, and audit logging. It will generate regulatory and performance reports such as grant utilization summaries, research output metrics, funding agency-specific reports, and publication impact statistics for accreditation and ranking purposes. Integration with external publication indexing services and the university's finance and HR systems is required. Due to the sensitivity of financial and intellectual property data, strong security, access control, versioning, and backup mechanisms must be implemented.

- 1.** i. Explain what type of project the above scenario represents. Is it **Organic**, **Semi-Detached** or **Embedded**? – Give proper reasoning.

ii. Find all the **COCOMO estimating parameters** for the project mentioned in the above scenario [show all calculations i.e., efforts, duration, productivity, budget etc]. Consider The average **line of code** required per function point is 40 and overall **Function Point** for the above scenario is 1250.

[CO3] 3+4 = 7

2. i. Define **Change Control** concept and mention its **importance** in software project management.
- ii. Explain the role of **Tester** in **Change control board (CCB)**.
- iii. Discuss the **steps** (basic course of events) and **alternate steps** for Change Control that should be followed in a software project management process.

[CO4] 2+1+4 = 7

3. i. Give a short brief on the concept **Dependencies** among the **Activities** for Project Scheduling in the perspective of building of a software.
- ii. Develop a **Gantt-chart** for the above project scenario by identifying the **tasks** and **tasks' dependency**. Also, mention different assumptions or considerations that need to be considered (e.g., weekly holidays and government holidays) for making a pragmatic estimation.

[CO5] 4+6 = 10

4. Which **Review method** will be appropriate for reviewing the project above, mention your justification. Also describe the **activities** of that review process.

[CO4] 2+4 = 6

5. Justify whether **DMAIC** or **DMADV** is more suitable for the **UIU Research Grant and Publication Management System**. Support your answer with reasons

[CO5] 5

6. Progress report submission and presentation on estimation through COCOMO model and Function point methods for a group project has been evaluated earlier also show the project management activities through the suitable PM Tool. (**Do not answer this question, already completed in classroom**).

[CO5] 5

Project Type	a	b	c	d
Organic	2.4	1.05	2.5	0.38
Semidetached	3.0	1.12	2.5	0.35
Embedded	3.6	1.20	2.5	0.30